

Ptbp1 is not required for retinal neurogenesis and cell fate specification

Reviewed Preprint

v2 • November 19, 2025

Revised by authors

Reviewed Preprint

v1 • September 19, 2025

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This study used a conditional knockout mouse line to remove Ptbp1 in retinal progenitors and demonstrated that its deletion has no effect on retinal neurogenesis or cell fate specification, thereby challenging the prevailing view of Ptbp1 as a master regulator of neuronal fate. The data are **convincing**, supported by transcriptomic analysis, histology, and proliferation assays. This study is **important**, and the broader implications for other CNS regions warrant further investigation.

<https://doi.org/10.7554/eLife.108331.2.sa3>

Abstract

The RNA-binding protein Ptbp1 has been proposed as a master regulator of neuronal fate, repressing neurogenesis through its effects on alternative splicing and miRNA maturation. While prior studies using RNA interference suggested that Ptbp1 loss promotes neurogenesis, recent genetic studies have failed to replicate glia-to-neuron conversion following *Ptbp1* loss of function. To evaluate the role of Ptbp1 in developmental neurogenesis *in vivo*, we conditionally disrupted *Ptbp1* in retinal progenitors. Ptbp1 was robustly expressed in both retinal progenitors and Müller glia but absent from postmitotic neurons, and efficient loss of function in mutant animals was confirmed using immunostaining for Ptbp1. Furthermore, bulk RNA-Seq at E16 revealed accelerated expression of late-stage progenitor and photoreceptor-specific genes and altered splicing patterns in *Ptbp1* mutants, including increased inclusion of rod photoreceptor-specific exons. However, we observed no defects in retinal lamination, progenitor proliferation, or cell fate specification in mature retina. ScRNA-Seq of mature mutant retinas revealed only modest transcriptional changes which partially recapitulate alterations seen following selective deletion of *Ptbp1* in mature glia. Our findings

demonstrate that *Ptbp1* is dispensable for retinal cell fate specification and suggest that its proposed role as a central repressor of neurogenesis should be reevaluated.

Introduction

Control of neurogenesis in progenitor cells of the developing CNS involves an elaborate interplay between negative and positive regulators. Neurogenic bHLH factors such as *Ascl1* and *Neurog2* induce neural progenitors to undergo terminal neurogenic divisions, while inhibitory factors such as components of the Notch pathway and NFI family transcription factors promote proliferative divisions and gliogenesis (Clark et al., 2019 [↗](#); Hufnagel et al., 2013 [↗](#); Keeley et al., 2023 [↗](#); Lyu et al., 2021 [↗](#); Yaron et al., 2006 [↗](#)). Studies in species such as zebrafish that undergo injury-induced neurogenesis by reprogramming of endogenous glial cells find that this process requires both upregulation of neurogenic factors and downregulation of inhibitory factors (Briona et al., 2015 [↗](#); Fausett et al., 2008 [↗](#); Lyu et al., 2023 [↗](#)). This has recently inspired many studies aimed at inducing neurogenic competence in mammalian glia by selectively targeting these factors (Hoang et al., 2020 [↗](#); Jorstad et al., 2017 [↗](#); Le et al., 2025, 2024; Todd et al., 2022 [↗](#), 2021).

One factor that has attracted considerable interest is the RNA binding protein and splicing regulator *Ptbp1*, which is widely expressed in non-neuronal cells, including both neural progenitors and glia (Boutz et al., 2007 [↗](#); Lilleväli et al., 2001 [↗](#)). It has been reported that *Ptbp1* acts as a broad inhibitor of neuronal-specific splicing (Boutz et al., 2007 [↗](#); Ling et al., 2016 [↗](#)), and inhibits maturation of miR-124a, which strongly promotes neurogenesis (Makeyev et al., 2007 [↗](#)), while its paralog *Ptbp2* promotes neuronal-specific splicing (Licatalosi et al., 2012 [↗](#); Li et al., 2014 [↗](#)). It was also proposed that *Ptbp1* acts as a master regulator of developmental neurogenesis in the CNS (Shibasaki et al., 2013 [↗](#)). This then inspired studies which used *Ptbp1* knockdown to attempt to reprogram glia into functional neurons. While it was reported that *Ptbp1* knockdown *in vivo* induced robust generation of functional retinal ganglion cells from Müller glia (Zhou et al., 2020 [↗](#)), as well as functional dopaminergic neurons from midbrain astrocytes (Qian et al., 2020 [↗](#)), attempts to replicate these results by other groups were unsuccessful (Chen et al., 2022 [↗](#); Hoang et al., 2023 [↗](#), 2022 [↗](#); Wang et al., 2021 [↗](#); Xie et al., 2022 [↗](#)). Notably, genetic loss of function of *Ptbp1* coupled with scRNA-Seq analysis failed to observe either loss of expression of glial-specific genes or induction of neuron-specific genes in mutant glia (Hoang et al., 2023 [↗](#), 2022 [↗](#)).

This result raises questions about previous claims about the central role of *Ptbp1* in controlling developmental neurogenesis. Previous functional studies of the role of *Ptbp1* in neuronal development were primarily conducted *in vitro* using RNAi to interrogate *Ptbp1* function (Linares et al., 2015 [↗](#); Spellman et al., 2007 [↗](#); Zheng et al., 2012 [↗](#)). The few *in vivo* studies of *Ptbp1* function in developing CNS have also almost exclusively used RNAi, and generally report modest and inconsistent phenotypes (Zhang et al., 2016 [↗](#)). Conventional *Ptbp1* homozygous mutants are lethal, precluding direct analysis of its role in developmental neurogenesis (Shibayama et al., 2009 [↗](#)). While one study reported accelerated cortical neurogenesis in *Ptbp1* heterozygotes (Shibasaki et al., 2013 [↗](#)), these effects were also quite modest and difficult to interpret, given the broader role of *Ptbp1* in early embryonic development.

The retina is a highly accessible and tractable model for the study of CNS neurogenesis and cell fate specification (Ptito et al., 2021 [↗](#)), and conditional mutagenesis allows selective loss of function of any gene of interest in retinal progenitors without affecting viability (Sauer, 1998 [↗](#)). We sought to use this approach to investigate the role of *Ptbp1* in regulating neurogenesis and cell fate specification in developing retina. While we observe changes in RNA splicing and a moderate acceleration of retinal development as measured by bulk RNA-Seq, we do not observe any changes in proliferation or neurogenesis during retinal development, or changes in the cell type

composition of *Ptbp1*-deficient retinas. Furthermore, we observe only modest changes in gene expression in adolescent mutant retina using scRNA-Seq analysis. These results demonstrate that *Ptbp1* is dispensable for retinal cell fate specification.

Results

***Ptbp1* mRNA is primarily expressed in retinal progenitors and Müller glia**

To examine the expression patterns of *Ptbp1* and *Ptbp2* in developing and adult neuroretina, we analyzed scRNA-Seq datasets from mouse and human retina that had been previously generated by our lab (Clark et al., 2019 [↗](#); Lu et al., 2020 [↗](#)). We observe high expression levels of *Ptbp1* mRNA in primary retinal progenitors in both species and Müller glia in mouse retina, with weaker expression in neurogenic progenitors, and little expression detectable in neurons at any developmental age. In contrast, *Ptbp2* is expressed in both progenitors and neurons, although with somewhat higher expression in retinal ganglion and amacrine cells (Fig. S1a [↗](#)). We observe that *Ptbp1* is expressed in primary progenitors at all developmental ages between E12 and P8 in mice and 9 and 20 gestational weeks in humans, although expression levels are moderately increased at later stages of neurogenesis (Fig. S1b [↗](#)).

***Chx10-Cre;Ptbp1^{lox/lox}* mice show loss of *Ptbp1* immunoreactivity in developing and mature retina**

To selectively disrupt *Ptbp1* function in retinal progenitors, we generated *Chx10-Cre;Ptbp1^{lox/lox}* mice. In these mice, Cre recombinase is expressed broadly in retinal progenitors from the beginning of neurogenesis onwards (Rowan and Cepko, 2004 [↗](#)), allowing efficient deletion of the conditional mutant allele of *Ptbp1* (Shibasaki et al., 2013 [↗](#)) (Fig. 1a [↗](#)). In some cases, the *Sun1-GFP* reporter line was used to visualize patterns of Cre activity (Mo et al., 2015 [↗](#)). Mutant mice were born at expected Mendelian ratios and showed no gross defects in retinal morphology. To confirm efficient deletion of *Ptbp1* in retinal progenitors, we performed immunohistochemistry for *Ptbp1* at E14, P1, and P30 in both control *Chx10-Cre;Ptbp1^{+/+}* and homozygous mutant and *Chx10-Cre;Ptbp1^{lox/lox}* and *Chx10-Cre;Ptbp1^{lox/lox};Sun1-GFP* mice (Fig. 1b-d [↗](#)). Perinuclear *Sun1-GFP* expression was observed in all retinal cells in *Chx10-Cre;Ptbp1^{lox/lox};Sun1-GFP* mice, reflecting the broad retinal progenitor-specific activity of the *Chx10-Cre* transgene, while nuclear GFP expression was observed in retinal progenitors (Mo et al., 2015 [↗](#)), reflecting the fact that a Cre-GFP fusion is used in the *Chx10-Cre* transgenic line (Rowan and Cepko, 2004 [↗](#)). At all stages, we observed a loss of *Ptbp1* protein expression in retinal progenitors and Müller glia, with a more than ten-fold reduction in the number of immunoreactive Müller glia observed at P30 (Fig. 1e [↗](#)). At both E14 and P1, we also observed a variable level of immunoreactivity in cells in the ganglion cell layer in control mice, which was also seen in *Chx10-Cre;Ptbp1^{lox/lox}* mice, suggesting that this may represent crossreactivity with an unknown epitope. This ganglion cell layer staining was not observed at P30, however. This demonstrates that *Ptbp1* is efficiently and selectively deleted in retinal progenitors from early stages of neurogenesis.

Embryonic *Chx10-Cre;Ptbp1^{lox/lox}* mice show shifts in temporal identity, increased expression of genes specific to photoreceptors, and altered patterns of RNA splicing

To further investigate the effects of *Ptbp1* deletion on embryonic progenitors, we conducted bulk RNA-Seq on whole retina from E16 wildtype, *Chx10-Cre;Ptbp1^{lox/+}*, and *Chx10-Cre;Ptbp1^{lox/lox}* retinas. A comparison of heterozygous *Chx10-Cre;Ptbp1^{lox/+}* and homozygous *Chx10-*

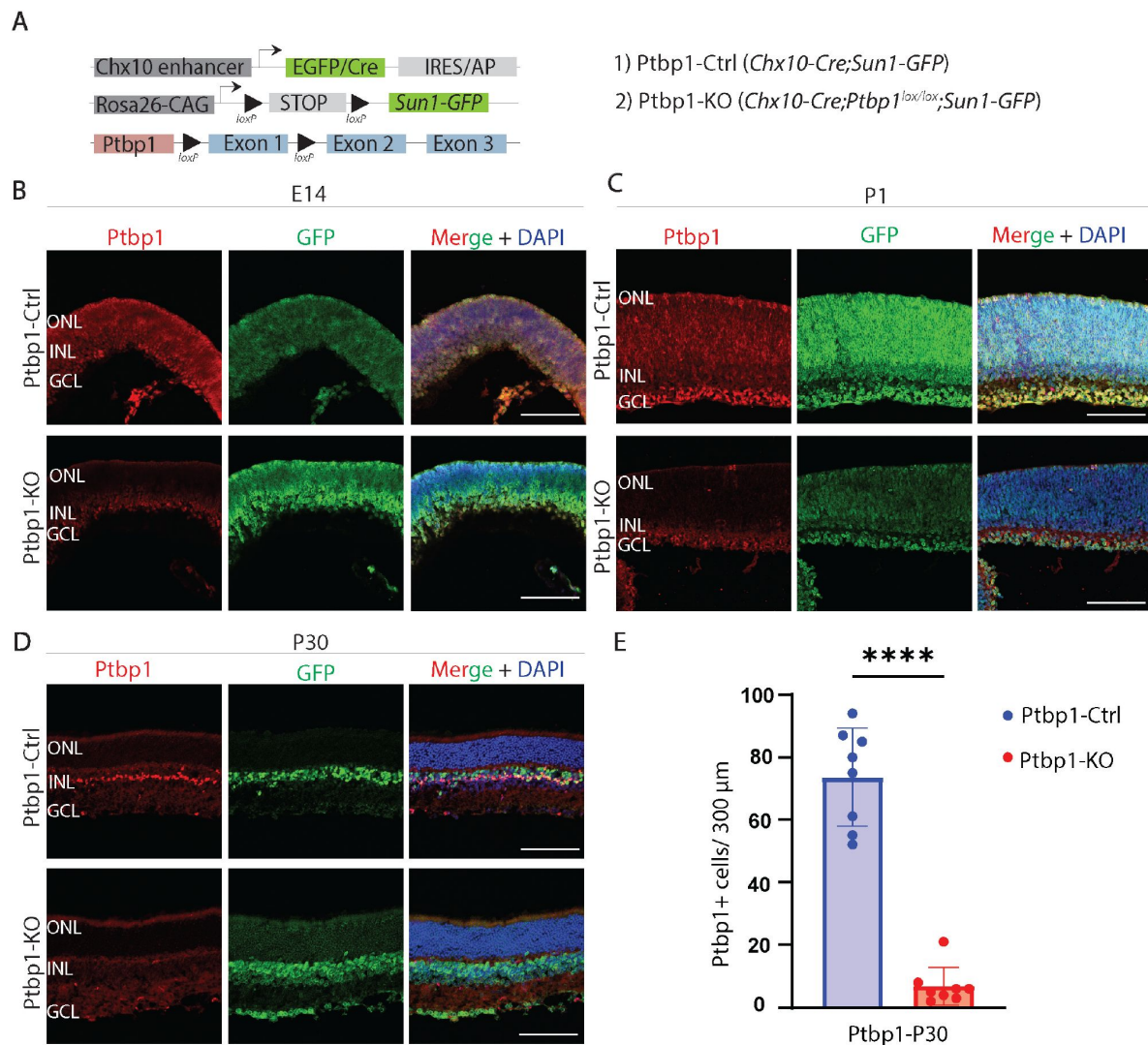


Figure 1.

***Ptbp1* is successfully deleted in *Chx10-Cre;Ptbp1^{lox/lox}* mice in developing and mature retina.**

(A) A schematic diagram of the generation of specific deletion of *Ptbp1* in early RPCs using *Chx10-Cre*. (B) Representative immunostaining for *Ptbp1* expression in *Chx10-Cre;Ptbp1^{+/+}* (Ptbp1-Ctrl) and *Chx10-Cre;Ptbp1^{lox/lox}* (Ptbp1-KO) mice at E14.5, (C) *Chx10-Cre;Ptbp1^{+/+};Sun1-GFP* and *Chx10-Cre;Ptbp1^{lox/lox};Sun1-GFP* mice at P1, (D) and *Chx10-Cre;Ptbp1^{+/+}* and *Chx10-Cre;Ptbp1^{lox/lox};Sun1-GFP* mice P30. (E) Quantification of *Ptbp1*-positive cells in Ptbp1-Ctrl and Ptbp1-KO retinas (n=8/genotype). Significance was determined via unpaired t test: **** p<0.0001. Each data point was calculated from an individual retina. ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar= 100 μm.

Cre;Ptbp1^{lox/lox} mutants identified 2075 genes that were significantly differentially expressed (Fig. 2a, Extended Data 1). As expected, homozygotes showed a loss of *Ptbp1* expression along with a compensatory upregulation of *Ptbp2*, as has previously been observed following selective disruption of *Ptbp1* in mature Müller glia (Hoang et al., 2022) (Fig. 2b). In addition, we observed a modest increase in expression of multiple genes specific to both rods and cone photoreceptors – including *Opn1sw* and *Rcvrn* – as well as altered expression of multiple genes selectively expressed in either early or late-stage RPCs (Clark et al., 2019; Lu et al., 2020; Lyu et al., 2021). For instance, the early-stage RPC-specific genes *Sfrp2* and *Foxp1* were downregulated, while late-stage RPC-enriched genes such as *Sox8*, *Nfix*, and *Hes5* were upregulated (Fig. 2b). Gene Ontology (GO) enrichment analysis likewise showed significant upregulation of genes involved in photoreceptor cell development and genes controlling synaptic function (Fig. 2c). However, immunohistochemical analysis does not detect any clear changes in the expression of HuC/D+NeuN, Otx2 or *Opn1sw* in *Chx10-Cre;Ptbp1^{lox/lox}* homozygous mutants relative to *Chx10-Cre;Ptbp1^{+/+}* control at either E14 or P0, and the number of Otx2-positive cells at P0 was not altered (Fig. S2). This implies that while at the transcriptional level retinal development in embryonic *Ptbp1* mutants appears to be moderately accelerated, based on higher expression of both markers of late-stage retinal progenitors and photoreceptors, this does not result in an increase in number of photoreceptors at this stage.

In light of the established role of *Ptbp1* in regulating RNA splicing in non-neuronal cells, we analyzed splicing differences in *Ptbp1*-deficient retinas using our dataset. We identified 864 differential percent-spliced-in (PSI) events, including changes in retained introns, alternative 5' or 3' splice site usage, mutually exclusive exons, and exon skipping, consistent with the known role of *Ptbp1* as a splicing repressor (Fig. 2d–e, Extended Data 1) (Vuong et al., 2016; Yap et al., 2012). To assess whether these splicing changes reflect neuronal or rod photoreceptor maturation, we compared our differential splicing events to those observed in adult neurons and rods (Ling et al., 2020). We found that 8 splicing events (7%) known to be enriched in central nervous system neurons and 19 events (35%) specific to rods overlapped with the splicing changes observed in *Ptbp1*-deficient retinas. (Fig. 2f, Extended Data 1).

Gene expression and splicing changes were compared across several reference tissues: heart tissue and Thy1-positive neurons, mature hair cells, microglia, and astrocytes (Fig. S3a,b). A heatmap of differentially expressed genes showed that while *Ptbp1*-deficient retinas diverged from WT retinas, their expression profiles did not resemble those of fully differentiated cell types like rods, astrocytes, or adult WT retina (Fig. S3c). Consistently, Pearson correlation analysis revealed that *Ptbp1*-deficient and WT retinas were more similar to each other than to fully differentiated neuronal or glial populations (Fig. S3d). Splicing profile analysis further revealed that while there was high correlation of PSI between *Ptbp1*-deficient and WT retinas, *Ptbp1*-deficient retinas more closely resembled Thy1-positive neurons, whereas WT retinas aligned more strongly with mature cells such as astrocytes, microglia, and auditory hair cells (Fig. S3e–f). Together, these results suggest that although *Ptbp1* loss of function induces hundreds of alternative splicing events, the magnitude of PSI changes in the KO retinas remains considerably lower than that seen in fully differentiated cell types (Extended Data 3). Thus, while a subset of splicing events overlaps with those characteristic of mature neurons or rods, the overall splicing and expression profiles of KO retinas are more similar to those of developing retinal tissue rather than terminally differentiated neuronal or glial populations.

***Chx10-Cre;Ptbp1^{lox/lox}* mice show no changes in retinal progenitor proliferation or neurogenesis**

To determine whether retinal progenitor-specific loss of function of *Ptbp1* resulted in a depletion of retinal progenitors, we conducted EdU labeling to quantify the number of cells in S-phase in P0 retina (Fig. 3a). We did not observe any significant differences in the number of EdU-positive cells between Cre negative *Ptbp1^{lox/lox}* controls and *Chx10-Cre;Ptbp1^{lox/lox}* mice (Fig. 3b,c). To

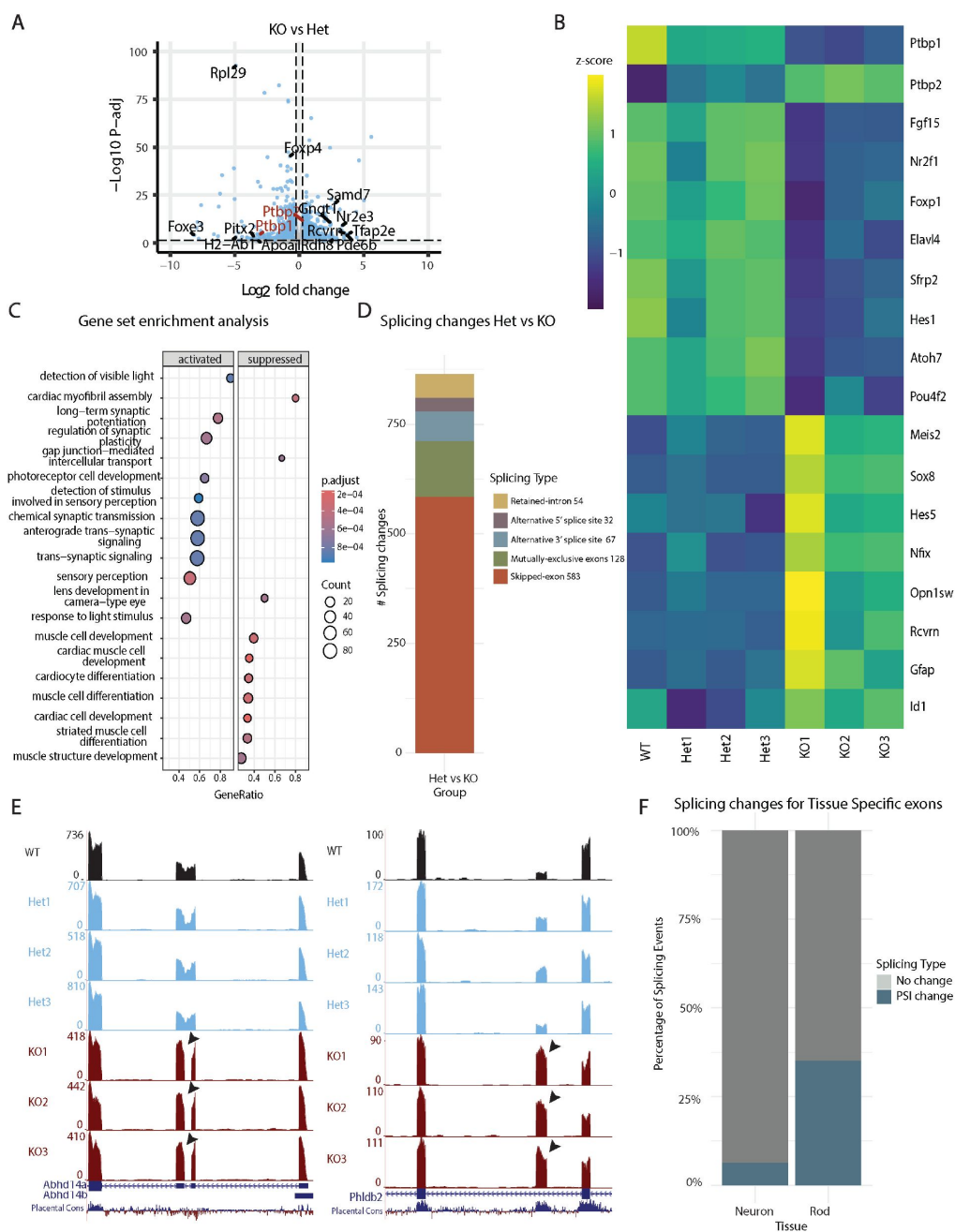


Figure 2.

BulkRNA-seq analysis reveals accelerated differentiation and altered RNA splicing in *Chx10-Cre;Ptbp1^{lox/lox}* mice at E16.5.

Volcano plot showing 2075 differentially expressed genes between heterozygous *Chx10-Cre;Ptbp1^{lox/+}* and homozygous *Chx10-Cre;Ptbp1^{lox/lox}* mutants as measured by bulk RNA-seq. Heatmap of select differentially expressed genes in wildtype *Chx10-Cre;Ptbp1^{+/+}*, heterozygous *Chx10-Cre;Ptbp1^{lox/+}*, and homozygous *Chx10-Cre;Ptbp1^{lox/lox}* samples. **(C)** Gene set enrichment analysis showing activated and repressed gene ontology terms in the homozygous *Ptbp1* mutant samples. **(D)** Barplot showing 864 splicing changes in homozygous *Ptbp1* mutant samples by splicing class including retained-introns (54 events), alternative 5' splice site (32 events), alternative 3' splice site (67 events), mutually-exclusive exons (128 events) and skipped-exons (583 events). **(E)** Track plots showing examples of alternative splicing of exons (arrows) in the *Ptbp1* mutant samples. **(F)** Bar plot showing that 7% of neuron-enriched and 35% of rod-specific splicing events overlap with splicing changes observed in *Ptbp1* mutants.

determine whether loss of function of *Ptbp1* influenced cell type specification in developing retina, we conducted immunostaining for a range of cell type-specific markers in *Chx10-Cre* control and *Chx10-Cre;Ptbp1^{lox/lox}* mice (Fig. 4). Although *Ptbp1* immunoreactivity was lost in homozygous mutant retinas (Fig. 4a), we did not observe any significant changes in the numbers or distribution of Sox9-positive Müller glia (Fig. 4b,f), HuC/D-positive amacrine and ganglion cells (Fig. 4c,f), Otx2-positive bipolar cells and photoreceptors (Fig. 4c,f), Rbpms-positive retinal ganglion cells (Fig. 4d,f), or cone arrestin-positive cone photoreceptors (Fig. 4e,f). Taken together, these findings show that *Ptbp1* loss of function induces neither precocious cell cycle exit nor affects normal levels and patterns of developmental neurogenesis and cell fate specification.

***Chx10-Cre;Ptbp1^{lox/lox}* mice show only modest changes in gene expression in the mature retina**

Despite the lack of any obvious defects in proliferation or cell fate specification in *Ptbp1*-deficient retinas, it nonetheless remained possible that this mutant might generate more subtle defects that are detectable using single-cell RNA-Seq analysis of whole retina. We therefore generated scRNA-Seq libraries from both P30 *Chx10-Cre;Ptbp1^{+/+}* and *Chx10-Cre;Ptbp1^{lox/lox}* mice. No changes in the identity or composition of any retinal cell type were observed (Fig. 5a-c). The overall identities and proportions of major retinal cell types appeared broadly similar between control and knockout sample (Fig. 5a-c). A modest number of genes in rod and cone photoreceptors, as well as bipolar neurons, showed statistically significant changes in gene expression (Fig. S4, Extended Data 2). A subset of phototransduction (*Pde6g*, *Guca1a*, *Pdc*, *Rcvrn*), photoreceptor-specific structural components (*Rom1*), and rod-specific transcription factors *Nrl* and *Nr2e3*, all showed modest but significant increases in expression in rods, while transcription factors such as *Otx2* and *Prdm1*, which are more prominently expressed in immature photoreceptors, showed significantly reduced expression. Cone photoreceptors showed increased expression of some phototransduction components (*Pde6c*, *Rcvrn*, *Guca1a*) and decreased expression of others (*Arr3*).

The largest number of differentially-expressed genes were observed in Müller glia (Fig. 5d). Several of the most strongly upregulated genes – notably including *Ptbp2*, *Fxyd6*, *Mt1*, and *Mt3* – were previously found to be upregulated by selective loss of function of *Ptbp1* in mature Müller glia in *Glast-CreER;Ptbp1^{lox/lox}* mice, with nearly a third of all significantly downregulated genes shared between both mutants (Hoang et al., 2022)(Fig. 5e, Extended Data 2). However, no significant changes in canonical markers of Müller glia such as *Sox9*, *Apoe*, *Glul*, or *Rlbp1* were observed (Fig. 5f).

Discussion

Previous studies have reported that *Ptbp1* acts in neural progenitors in the CNS as a master repressor of neuronal fate (Hu et al., 2018; Shibasaki et al., 2013). If this were the case, we would expect that *Ptbp1* loss of function in early development would rapidly lead to precocious cell cycle exit and reduction in the number of progenitor cells. As a result, we would expect to observe an overrepresentation of early-born cell types such as retinal ganglion cells and cone photoreceptors at the expense of late-born cell types such as bipolar interneurons and, in particular, Müller glia. However, we observe that while *Ptbp1* is robustly expressed in both retinal progenitors and mature glia, as in other CNS regions, loss of *Ptbp1* function in early-stage retinal progenitors does not induce any significant developmental defects. Despite efficient and widespread loss of *Ptbp1* expression from E14 onwards, we do not observe any changes in progenitor proliferation, retinal cell type composition, or expression of canonical Müller glia marker genes in mature retina. The changes in gene expression that are observed in adolescent *Ptbp1*-deficient retinas are quite modest, although increased levels of several photoreceptor-

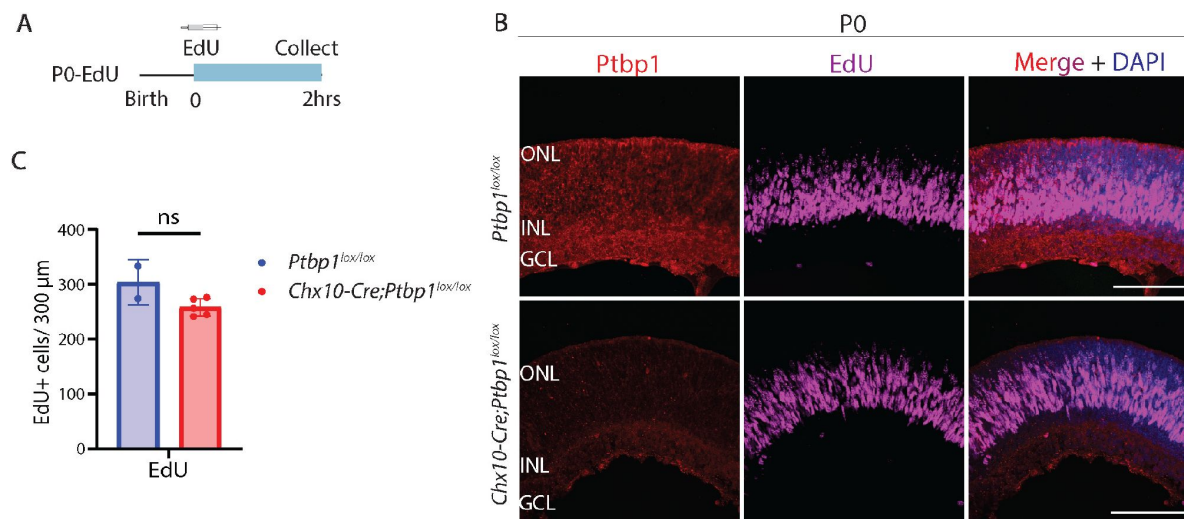


Figure 3.

***Ptbp1* deletion does not alter retinal progenitor proliferation.**

(A) A schematic diagram of the experimental timeline for labeling dividing cells using EdU. **(B)** Representative immunostaining for Ptbp1 expression and EdU labeling in Ptbp1-Ctrl and Ptbp1-KO mice at P0. **(C)** Quantification of EdU-positive cells in Ptbp1-Ctrl and Ptbp1-KO retinas ($n \geq 2$ /genotype). Significance was determined via Mann-Whitney test: $ns = p > 0.05$. Each data point was calculated from an individual retina. ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar=100 μ m.

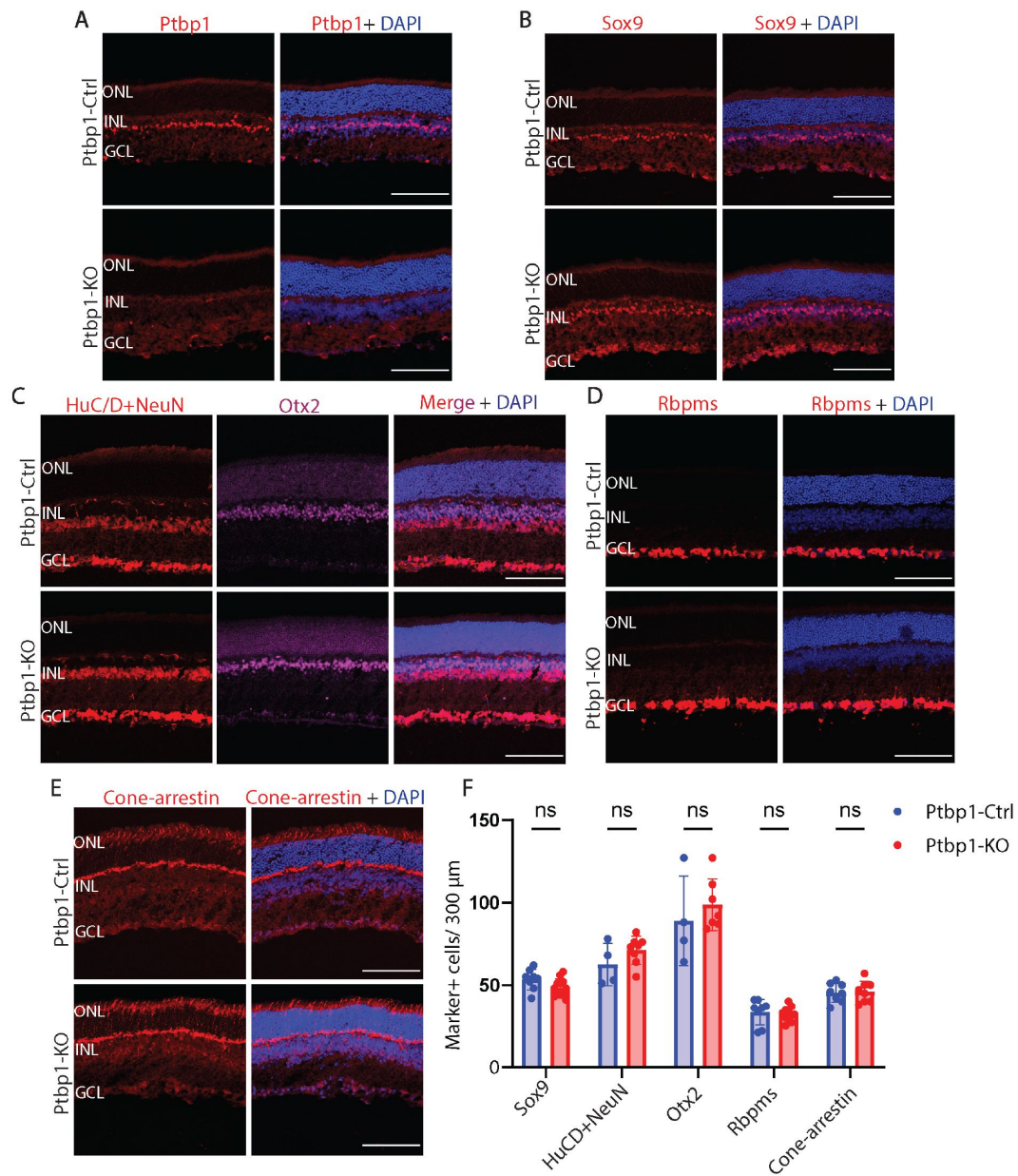


Figure 4.

***Ptbp1* deletion does not alter retinal neurogenesis.**

(A) Representative immunostaining for *Ptbp1*, (B) *Sox9*, (C) *HuC/D+NeuN*, *Otx2*, (D) *Rbpms*, and (E) *Cone-arrestin* in *Ptbp1*-Ctrl and *Ptbp1*-KO mice at P30. (F) Quantification of *Sox9*, *HuC/D+NeuN*, *Otx2*, *Rbpms*, and *Cone-arrestin*-positive cells in *Ptbp1*-Ctrl and *Ptbp1*-KO retinas ($n \geq 4$ /genotype). Significance was determined via multiple unpaired *t* tests: ns= $p > 0.05$. Each data point was calculated from an individual retina. ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar= 100 μ m.

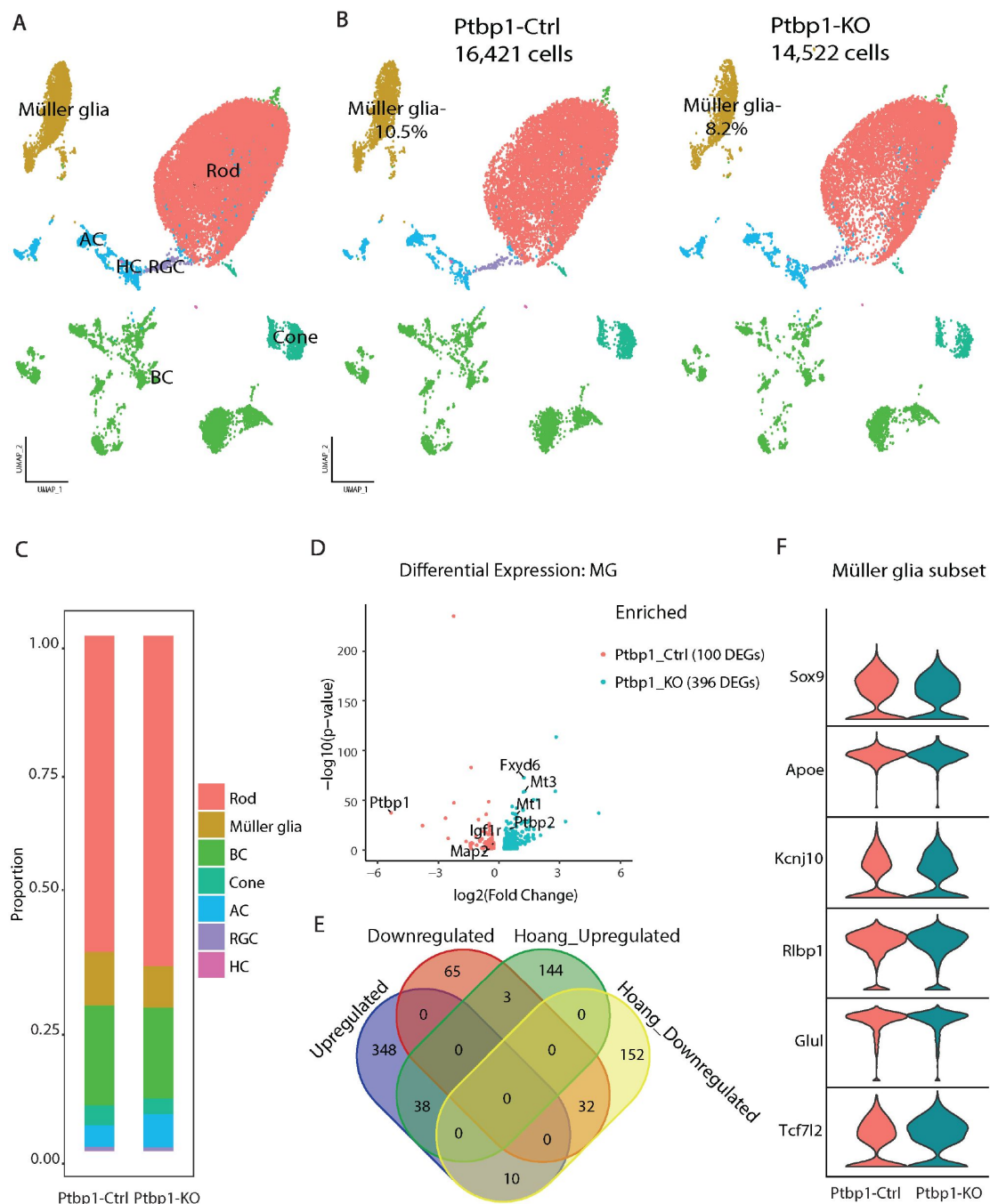


Figure 5.

scRNA-seq analysis reveals modest transcriptional changes in Müller glial gene expression in *Chx10-Cre;Ptbp1^{lox/lox}* mice.

(A) Uniform manifold approximation and projection (UMAP) plots showing cell types captured in the *Chx10-Cre;Ptbp1^{+/+}* (Ptbp1-Ctrl) and *Chx10-Cre;Ptbp1^{lox/lox}* (Ptbp1-KO) merged dataset. (B) UMAP plots showing cell types separated by genotype with the percentage of Müller glia in each group. (C) Barplot showing cell proportions for each cell type in both genotypes. (D) Volcano plot showing differentially expressed genes in Müller glia between Ptbp1-Ctrl and Ptbp1-KO samples. (E) Venn diagram comparison of upregulated and downregulated differentially expressed genes for Müller glia between this dataset and previously published data analyzing Müller glia-specific loss of function of *Ptbp1* (Hoang et al., 2022). (F) Violin plot showing no significant changes in the expression of canonical Müller glia markers in Ptbp1-KO sample compared to Ptbp1-Ctrl.

specific genes are observed. This is consistent with previous studies showing that *Ptbp1* knockdown enhances photoreceptor-specific splicing of a subset of alternatively spliced genes (Ling et al., 2020 [DOI](#)), and implies that this may indirectly increase expression of a subset of rod-specific genes.

At E16, however, homozygous *Ptbp1* mutants do show changes in gene expression consistent with previously reported functions. These retinas show increased expression of *Ptbp2*, the neuronally-enriched paralog of *Ptbp1*, along with increased expression of genes specific to late-stage progenitors and photoreceptor precursors. Furthermore, *Ptbp1* mutants exhibit altered RNA splicing and inclusion of a subset of exons specific to rod photoreceptors. While this is apparently consistent with reports that developmental loss of *Ptbp1* function promotes neurogenesis by inducing a neuronal-like splicing pattern in neural progenitors (Boutz et al., 2007 [DOI](#); Hu et al., 2018 [DOI](#); Shibasaki et al., 2013 [DOI](#); Vuong et al., 2016 [DOI](#)), this does not result in changes in either progenitor proliferation or any change in retinal cell type composition in either the developing or adolescent retina. This argues against the idea that *Ptbp1* loss simply accelerates the timing of neurogenesis, as there is no evidence for increased photoreceptor production or altered proliferation in embryonic or neonatal retina. An alternative interpretation would be that *Ptbp1* loss of function leads to the precocious formation of ectopic neurons which rapidly undergo cell death. Further studies will be needed to resolve this question.

Following closely behind studies that fail to show any role for *Ptbp1* in regulating glia-to-neuron reprogramming in adult CNS (Chen et al., 2022 [DOI](#); Hoang et al., 2023 [DOI](#), 2022 [DOI](#); Wang et al., 2021 [DOI](#)), these findings call into question previous models that place *Ptbp1* in a central position in controlling developmental neurogenesis. The reason for this discrepancy is not entirely clear. Previous studies using knockdown may have been complicated by off-target effects (Jackson et al., 2003 [DOI](#)), and conditions for *in vitro* analysis may not have accurately replicated conditions in the native CNS. It is also possible that compensatory mechanisms differ between knockdown and knockout approaches. Notably, a recent study (Konar et al., 2025 [DOI](#)) reported that *Ptbp1* knockdown promotes Müller glia proliferation in zebrafish, suggesting that effects of acute reduction of *Ptbp1* mRNA may not fully mirror those of complete loss-of-function. Alternatively, upregulation of the widely expressed paralog *Ptbp2* may have compensated for developmental defects resulting from *Ptbp1* loss of function. Generation of conditional double mutant mice will be necessary to definitively exclude this possibility. Additionally, the use of *Chx10-Cre* in this study does not exclude a potential role for *Ptbp1* during early stages of retinal development prior to E11, when the Cre is first expressed. Regardless, this study demonstrates that *Ptbp1* is dispensable for the process of retinal cell fate specification, and we anticipate that this will likely hold for other CNS regions.

Methods

Mice

Mice were raised and housed in a climate-controlled pathogen-free facility on a 14/10 h light/dark cycle. Mice used in this study were *Chx10-Cre* (JAX #005105) *Ptbp1*^{lox/lox} mice carrying loxP sites that flank the promoter and 1st exon of *Ptbp1* were generated as described previously (Shibayama et al., 2009). *Chx10-Cre;Ptbp1*^{lox/lox} mice were generated by crossing *Chx10-Cre* with conditional *Ptbp1*^{lox/lox} mice. *Chx10-Cre;Ptbp1*^{lox/lox}; *Sun1-GFP* mice were also examined in some cases to visualize Cre activity in postmitotic neurons (Mo et al., 2015 [DOI](#)). Maintenance and experimental procedures performed on mice were in accordance ARRIVE guidelines for animal research, with the protocol approved by the Institutional Animal Care and Use Committee (IACUC) at the Johns Hopkins School of Medicine under protocol number MO22M22.

EdU treatment

Chx10-Cre control and *Chx10-Cre;Ptbp1^{lox/lox}* neonatal mice were administered EdU (Abcam, #ab146186) via subcutaneous injection at P0 [10–20 µl of 5 mg/ml in PBS]. Retinas were collected two hours post EdU injection for analysis.

Immunohistochemistry and imaging

Collection and immunohistochemical analysis of retinas were performed as described previously (Hoang et al., 2020 [DOI](#)). Briefly, mouse eye globes were fixed in 4% paraformaldehyde (ElectronMicroscopySciences, #15710) for 1 h at 4°C. Retinas were dissected in 1x PBS and incubated in 30% sucrose overnight at 4°C.

Retinas were then embedded in OCT (VWR, #95057-838), cryosectioned at 16 µm thickness, and stored at –20°C. Sections were dried for 30 min in a 37°C incubator and washed 3 x 5 min with 0.1% TritonX-100 in PBS (PBST). EdU labeling was performed by using Click-iT EdU kit (ThermoFisher, #C10340) following the manufacturer's instructions. Sections were then incubated in 10% Horse Serum (ThermoFisher, #26050070), 0.4% TritonX-100 in 1x PBS (blocking buffer) for 2 h at room temperature (RT) and then incubated with primary antibodies in the blocking buffer overnight at 4°C. Primary antibodies used in this study were: rabbit anti-Ptbp1 (1:200, Invitrogen, #PA3-81297), rabbit anti-Ptbp1 (1:200, ABclonal, #A6107), rabbit anti-RBPMS (1:200, Proteintech, #151871-AP), rabbit anti-cone arrestin (1:200, Millipore Sigma, #AB15282), goat anti-OTX2 (1:200, R&D Systems, #AF1979), mouse anti-HuC/D (1:200, ThermoFisher Scientific, #A21271), mouse anti-NeuN (1:200, SigmaAldrich, #MAB377), rabbit anti-SOX9 (1:200, SigmaAldrich, #AB5535) and chicken anti-GFP (1:400, ThermoFisher Scientific, #A10262). An equal mixture of anti-HuC/D and NeuN antibodies was used to detect amacrine and ganglion cells.

Sections were washed 3 x 5 min with PBST to remove excess primary antibodies and were incubated in secondary antibodies in blocking buffer for 2 h at RT. Sections were then counterstained with DAPI in PBST, washed 3 x 5 min in PBST and mounted with ProLong Gold Antifade Mountant (Invitrogen, #P36935) under coverslips (VWR, #,48404-453), air-dried, and stored at 4°C. Fluorescent images were captured using a Zeiss LSM 700 confocal microscope.

Cell quantification and statistical analysis

Ptbp1-positive, Sox9-positive, HuC/D/Neun-positive, Otx2-positive, Rbpms-positive, and ArrestinC-positive cells were counted in total from 300µm sections per retina. For cell proliferation quantification, EdU-positive cells were counted from 300µm sections per retina. Each data point in the bar graphs was calculated from an individual retina. All cell quantification data were graphed and analyzed using GraphPad Prism 10. Multiple unpaired *t* tests or Mann-Whitney tests were used for analysis between samples of multiple groups. In all tests, values of *p* < 0.05 were considered to indicate significance.

Retinal cell dissociation

Briefly, one female mouse per genotype was euthanized by CO₂. Retinas were dissected in fresh ice-cold PBS and retinal cells were dissociated using Papain Dissociation System (LK003150, Worthington). Dissociated cells were resuspended in ice-cold HBAG buffer containing 9.8mL HibernateA (BrainBits, #HALF500), 200mL B27 (ThermoFisher, #17504044), 20mL GlutaMAX (ThermoFisher, #35050061) and 0.5U/mL RNase inhibitor. Cells were filtered through a 50mm filter. Cell count and viability were determined by using 0.4% Trypan blue.

Bulk RNA-sequencing library preparation

Retinas were dissected from E16 mice embryos and stored at -80°C. Frozen retinal tissue was pooled into 7 samples by genotype: wildtype (WT (6 pooled retinas)); heterozygous *Ptbp1* mutant *Chx10-Cre;Ptbp1^{lox/+}* (Het1 (5 pooled retinas), Het2 (5 pooled retinas), Het3 (6 pooled retinas)); and homozygous *Ptbp1* mutant (KO1 (4 pooled retinas), KO2 (4 pooled retinas), and KO3 (6 pooled retinas)). RNA was extracted using the RNeasy Micro Plus kit (Qiagen). Illumina Stranded mRNA Prep Ligation kit was used for bulk RNA library preparation. Libraries were sequenced on the Illumina NovaSeq X.

Single-cell RNA-sequencing (scRNA-seq) library preparation

scRNA-seq were prepared on dissociated retinal cells using the 10X Genomics Chromium Single Cell 3' Reagents Kit v3.1 (10X Genomics, Pleasanton, CA). Libraries were constructed following the manufacturer's instructions and were sequenced using the Illumina NovaSeq 6000. The sequencing data were aligned to the mm10 reference genome and a cell-by-gene matrix was generated using the Cell Ranger 7.0.1 pipeline (10X Genomics).

ScRNA-seq data analysis

ScRNA-seq of control *Chx10-Cre;Ptbp1^{+/+}* and homozygous mutant *Chx10-Cre;Ptbp1^{lox/lox}* mice were analyzed using the Seurat package (Hao et al., 2021 [↗](#)). Briefly, cells were filtered to retain those with more than 200 and fewer than 5,000 detected genes, more than 700 and fewer than 10,000 total transcripts, and less than 20% mitochondrial gene content. The data were normalized using the default log-normalization parameters, the top 4000 highly variable features were identified, and data were scaled while regressing out mitochondrial gene content. Principal component analysis (PCA) was then performed on the variable genes followed by louvain clustering and UMAP visualization using the top 12 principal components. Expression of key marker genes within each cell cluster were used to confirm the appropriate assignment of cell types. Differentially expressed genes between groups were determined using the Wilcoxon rank sum test. The clusterProfiler R package was used to perform gene ontology enrichment analysis of biological processes found in the differentially expressed gene lists (Yu et al., 2012 [↗](#)).

Bulk RNA-seq analysis

Reads were processed using the rMATS-turbo pipeline (Wang et al., 2024 [↗](#)), which performs alignment with STAR and custom splice junction counts. Alignments were generated using the GRCh38 mouse genome (Release M10) with GENCODE annotation. Gene-level counts were then extracted from the resulting BAM files using featureCounts, and differential gene expression was analyzed with the DESeq2 package in R (Liao et al., 2014 [↗](#); Love et al., 2014 [↗](#)). Comparisons between *Ptbp1* heterozygous (Het) and knockout (KO) samples identified differentially expressed genes based on an absolute log2 fold change greater than 0.263 (representing at least a 20% change in expression) and an adjusted p-value below 0.05. We incorporated publicly available bulk RNA-seq datasets for comparative analysis, including embryonic day 16 retina (E16: GSM2720095, GSM2720096), postnatal day 28 retina (P28: GSM2720111, GSM2720112), Thy1-positive neurons (GSM2392791), astrocytes (GSM1269903, GSM1269904), microglia (GSM1269913, GSM1269914), hair cells (GSM1602228, GSM1602229), and heart tissue (GSM1223635) (Brooks et al., 2019 [↗](#); Cai et al., 2015 [↗](#); Giudice et al., 2014 [↗](#); Yang et al., 2017 [↗](#); Zhang et al., 2014 [↗](#)). Hair cells and heart tissue samples were chosen as examples of high and low *Ptbp1* expression, respectively, based on data from ASCOT (Ling et al., 2020 [↗](#)). We generated a Pearson correlation matrix to assess overall sample similarity. Splicing differences were evaluated using filtered rMATS output, applying the following thresholds: read coverage ≥ 15 , minimum PSI = 0.05, maximum PSI = 0.95, significant FDR (sigFDR) ≤ 0.05 , $|\Delta\text{PSI}| \geq 0.10$, background FDR (bgFDR) ≤ 0.5 , and background within-group $\Delta\text{PSI} \leq 0.2$. For the E16 and P28 retina datasets, we used more relaxed filters (read coverage ≥ 0 , PSI between 0.01 and 0.99) to ensure inclusion of lower-

abundance events. Splicing events specific to rods and various neuronal subtypes were queried from the ASCOT supplementary files and visually inspected using the UCSC Genome Browser to assess their presence in bulk RNA-seq datasets. This analysis followed the approach described in previous studies (Carmen-Orozco et al., 2024 [DOI](#); Ling et al., 2020 [DOI](#)).

Supplementary figures

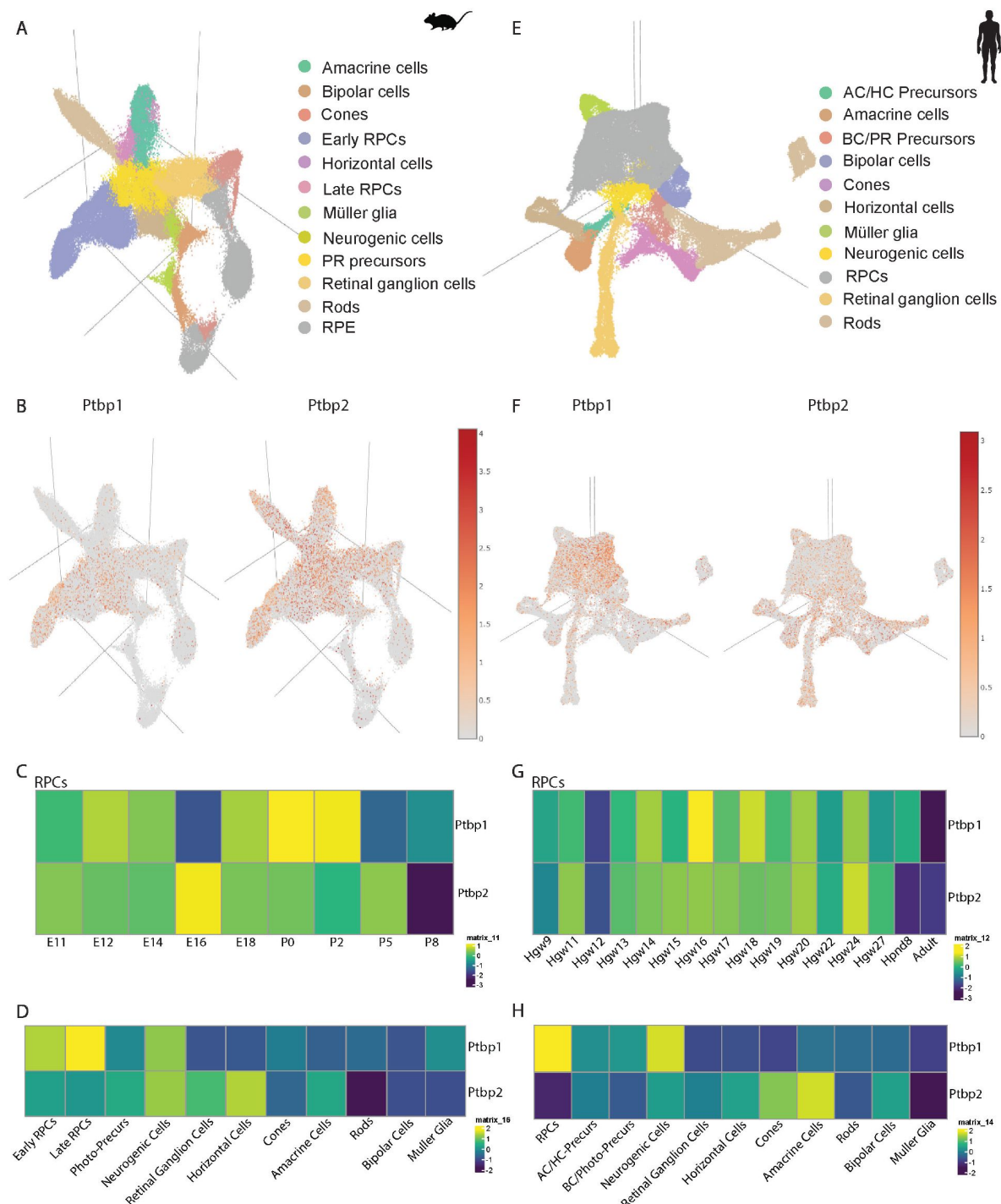


Figure S1.

Cellular expression patterns of *Ptbp1* and *Ptbp2* during mouse and human retinal development.

(A) 3D UMAP plot showing cell types from combined timepoints (E14-Adult) of mouse retinal development utilizing publicly available data from (Clark et al., 2019). (B) Gene plots for *Ptbp1* and *Ptbp2* expression during mouse retinal development. (C) Heatmaps of *Ptbp1* and *Ptbp2* expression patterns in RPCs at varying time points during mouse retinal development. (D) Heatmaps of *Ptbp1* and *Ptbp2* expression patterns in different mouse retinal cell types. (E) 3D UMAP plot showing cell types from combined timepoints (Hgw9-Adult) of human retinal development utilizing publicly available data from (Lu et al., 2020). (F) Gene plots for *Ptbp1* and *Ptbp2* expression during human retinal development. (G) Heatmaps of *Ptbp1* and *Ptbp2* expression patterns in RPCs at varying time points during human retinal development. (H) Heatmaps of *Ptbp1* and *Ptbp2* expression patterns in different human retinal cell types.

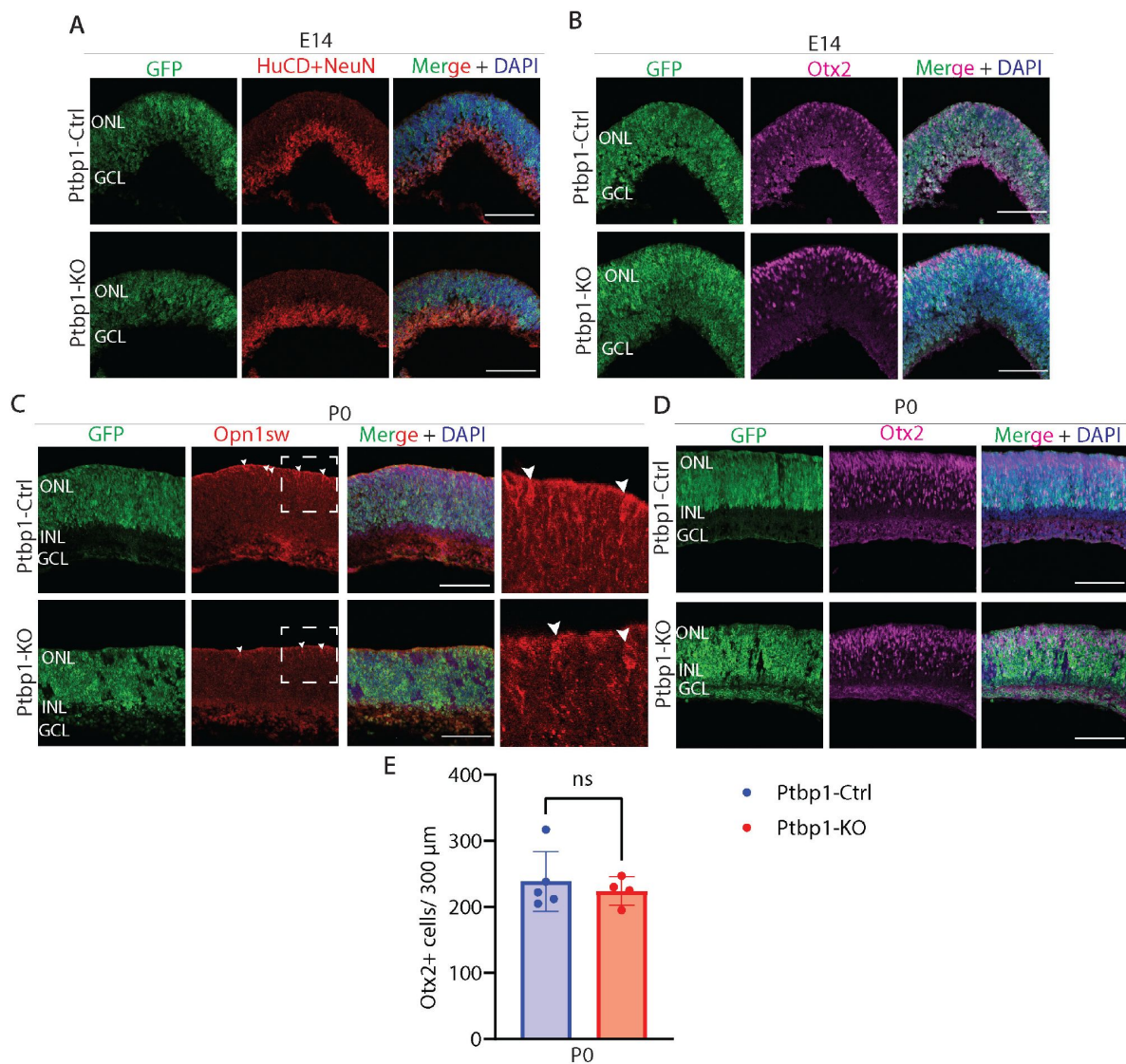


Figure S2.

Immunostaining does not detect changes in cell composition in E14 or P0 *Ptbp1*-deficient retina.

(A) Representative immunostaining for, HuC/D+NeuN and **(B)** Otx2 in *Ptbp1*-Ctrl and *Ptbp1*-KO mice at E14. **(C)** Representative immunostaining for Opn1sw and **(D)** Otx2 in *Ptbp1*-Ctrl and *Ptbp1*-KO mice at P0, with the *Ptbp1*-KO mouse having the *Chx10-Cre;Ptbp1^{lox/lox};Sun1-GFP* genotype (Mo et al., 2015). Note that GFP signal is found in all retinal cell types, since the Cre transgene is active in retinal progenitors over the full course of neurogenesis. **(E)** Quantification of Otx2-positive cells in *Ptbp1*-Ctrl and *Ptbp1*-KO retinas at P0 ($n \geq 4$ /genotype). Significance was determined via Mann-Whitney test: ns= $p > 0.05$. Each data point was calculated from an individual retina. ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar= 100 μ m.

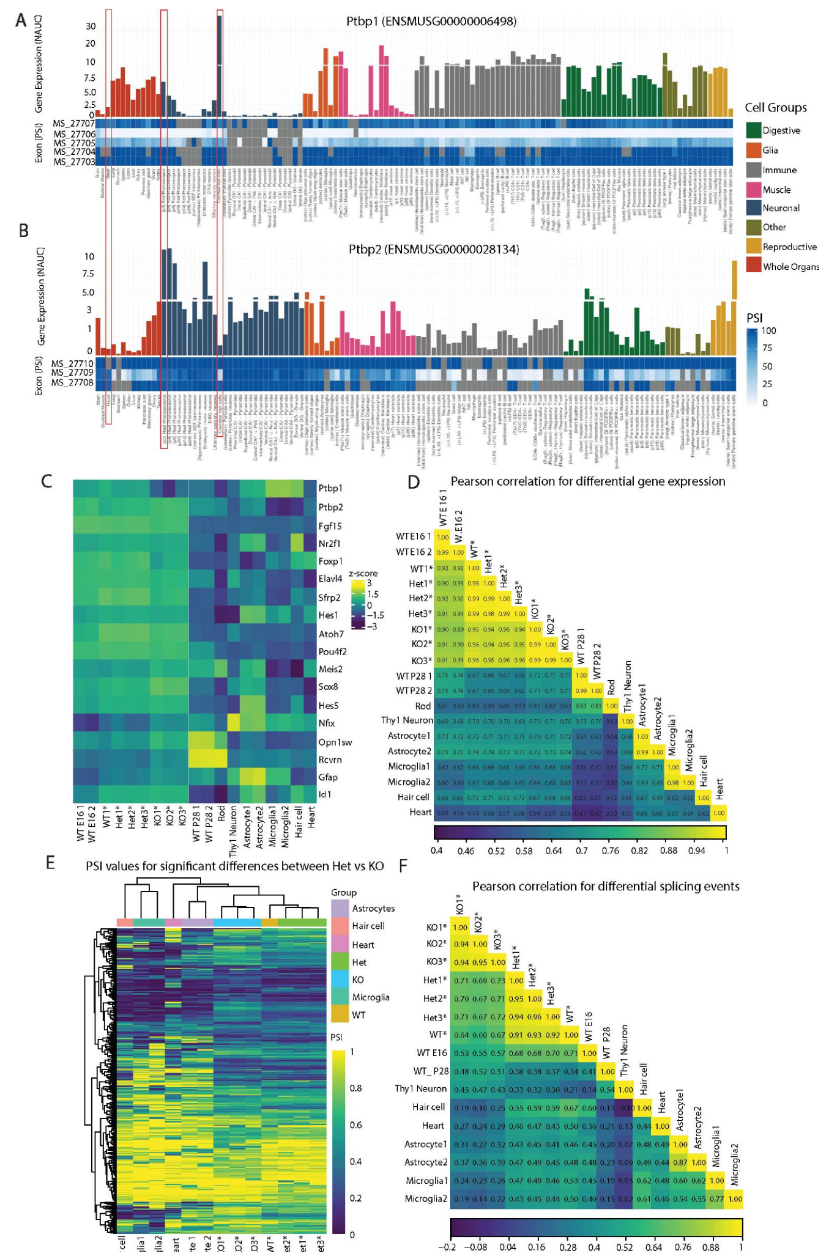


Figure S3.

BulkRNA-seq analysis reveals altered gene expression and splicing patterns in E16 *Chx10-Cre;Ptbp1^{lox/lox}* mice retina.

(A) Bar charts showing *Ptbp1* and (B) *Ptbp2* expression in various cell types and tissues using the ASCOT browser (Ling et al., 2020). Highlighted samples are rods photoreceptors at P2, heart tissue showing low *Ptbp1* expression and cochlear hair cells showing high *Ptbp1* expression. (C) Heatmap of select differentially expressed genes between individual replicates samples for wildtype (WT), heterozygous *Chx10-Cre;Ptbp1^{lox/+}* (Het1, Het2, Het3), and homozygous *Chx10-Cre;Ptbp1^{lox/lox}* (KO1, KO2, KO3) *Ptbp1* mutant retinal samples analyzed in this study, along with publicly available bulk RNA-seq datasets including embryonic day 16 retina, postnatal day 28 retina, Thy1-positive neurons, astrocytes, microglia, hair cells and heart tissue. (D) Pearson correlation plot for identified differentially expressed genes between Het and KO samples, plotted across all samples including publicly available datasets used. (E) Heatmap showing PSI values and (F) pearson correlation plot for splicing events identified as differentially spliced between Het and KO samples, plotted across all samples including publicly available datasets.

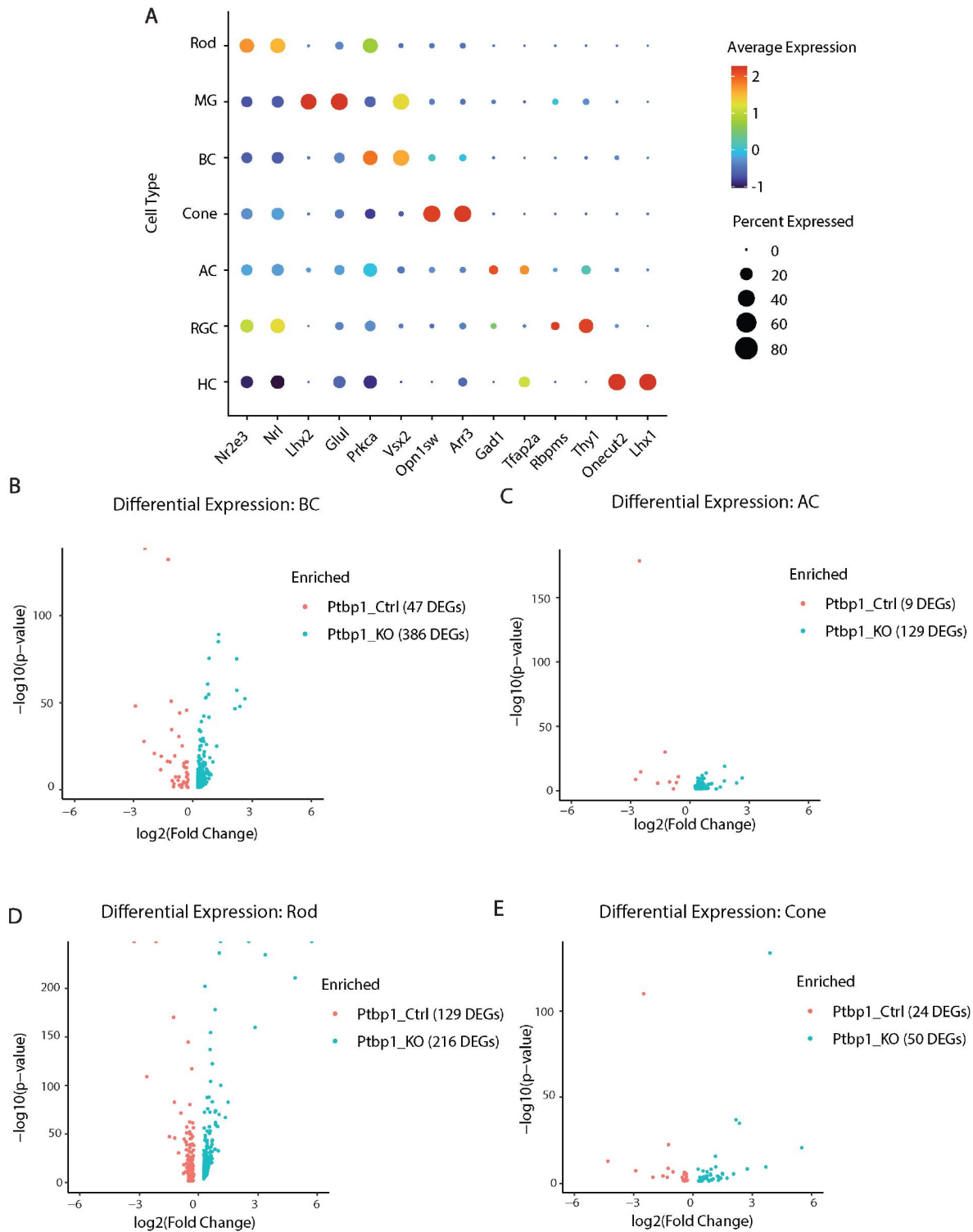


Figure S4.

ScRNA-seq analysis identifies differentially expressed genes in major adolescent retinal cell types following developmental loss of function of *Ptbp1*.

(A) Dot Plot of cell type-specific markers used to identify clusters for each major retinal cell type. (B) Volcano plot showing differentially expressed genes for bipolar cells (BC), (C) amacrine cells (AC), (D) rod photoreceptors and (E) cone photoreceptors between Ptbp1-Ctrl and Ptbp1-KO samples.

Acknowledgements

This work was supported by NIH grant R01EY036173 to S.B.

Additional information

Data availability

All raw RNA-Seq and scRNA-Seq data are publicly available through the Gene Expression Omnibus under accession numbers GSE300588 and GSE300607. Analysis scripts used in this study are accessible at: https://github.com/csanti88/ptbp1_2025. Bulk RNA-seq and other tissue track plots visualizations are available at: https://genome.ucsc.edu/s/roggercarmen/Ptbp1KO_Retina

Funding

National Eye Institute (NEI) (R01EY036173)

- Seth Blackshaw

Additional files

Extended Data 1. Bulk RNA-Seq analysis. Table of 2075 differentially expressed genes between heterozygous *Chx10-Cre;Ptbp1^{lox/+}* and homozygous *Chx10-Cre;Ptbp1^{lox/lox}* mutants identified by bulk RNA-Seq at E16.5, table of 864 differential splicing changes, and table of adult neuronal and rod enriched splicing events that overlap with the differential splicing events in the bulk RNA-Seq dataset.

Extended Data 2. scRNA-Seq analysis. Proportions of cell types in the *Ptbp1*-Ctrl and *Ptbp1*-KO scRNA-Seq datasets. Table of differentially expressed genes between *Chx10-Cre;Ptbp1^{+/+}* control and homozygous *Chx10-Cre;Ptbp1^{lox/lox}* mutants identified by scRNA-Seq in adolescent retina for Müller glia (496), bipolar cells (433), amacrine cells (138), rod photoreceptor (345) and cone photoreceptor cells (74) with adjusted *p*-value of 0.05. Positive *avg_log2FC* scores represent upregulated genes within the *Ptbp1*-KO sample.

Extended Data 3. Bulk RNA-Seq analysis of Rod and Neuron-specific exons. Genome-browser view showing exon-level read coverage for 18 Rod-specific (A) and 8 Neuron-specific (B) exons across WT, heterozygous *Chx10-Cre;Ptbp1^{lox/+}* and homozygous *Chx10-Cre;Ptbp1^{lox/lox}* mutant retinas and additional tissue including brain, rod photoreceptors (P2, P6, P14, P28), and various neuronal and glial cell types. Tracks indicate RNA-seq read density aligned to the mm10 genome, highlighting exon-specific differences in expression and potential alternative splicing events across genotypes.

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Reviewer #1 (Public review):

Summary:

The researchers sought to determine whether Ptbp1, an RNA-binding protein formerly thought to be a master regulator of neuronal differentiation, is required for retinal neurogenesis and cell fate specification. They used a conditional knockout mouse line to remove Ptbp1 in retinal progenitors and analyzed the results using bulk RNA-seq, single-cell RNA-seq, immunohistochemistry, and EdU labeling. Their findings show that Ptbp1 deletion has no effect on retinal development, since no defects were found in retinal lamination, progenitor proliferation, or cell type composition. Although bulk RNA-seq indicated changes in RNA splicing and increased expression of late-stage progenitor and photoreceptor genes in the mutants, and single-cell RNA-seq detected relatively minor transcriptional shifts in Müller glia, the overall phenotypic impact was low. As a result, the authors conclude that Ptbp1 is not required for retinal neurogenesis and development, thus contradicting prior statements about its important role as a master regulator of neurogenesis. They argue for a reassessment of this stated role. While the findings are strong in the setting of the retina, the larger implications for other areas of the CNS require more investigation. Furthermore, questions about potential reimbursement from Ptbp2 warrant further research.

Strengths:

This study calls into doubt the commonly held belief that Ptbp1 is a critical regulator of neurogenesis in the CNS, particularly in retinal development. The adoption of a conditional knockout mouse model provides a reliable way for eliminating Ptbp1 in retinal progenitors while avoiding the off-target effects often reported in RNAi experiments. The combination of bulk RNA-seq, scRNA-seq, and immunohistochemistry enables a thorough examination of molecular and cellular alterations at both embryonic and postnatal stages, which strengthens the study's findings. Furthermore, using publicly available RNA-Seq datasets for comparison improves the investigation of splicing and expression across tissues and cell types. The work is well-organized, with informative figure legends and supplemental data that clearly show no substantial phenotypic changes in retinal lamination, proliferation, or cell destiny, despite identified transcriptional and splicing modifications.

Weaknesses:

The retina-specific method raises questions regarding whether Ptbp1 is required in other CNS locations where its neurogenic roles were first proposed. Although the study performs well in transcriptome and histological analyses, it lacks functional assessments (such as electrophysiological or behavioral testing) to determine if small changes in splicing or gene expression affect retinal function.

<https://doi.org/10.7554/eLife.108331.2.sa2>

Reviewer #2 (Public review):

Summary:

Ptbp1 has been proposed as a key regulator of neuronal fate through its role in repressing neurogenesis. In this study, the authors conditionally inactivated Ptbp1 in mouse retinal progenitor cells using the Chx10-Cre line. While RNA-seq analysis at E16 revealed some changes in gene expression, there were no significant alterations in retinal cell type composition, and only modest transcriptional changes in the mature retina, as assessed by immunofluorescence and scRNAseq. Based on these findings, the authors conclude that Ptbp1 is not essential for cell fate determination during retinal development.

Strengths:

Despite some effects of Ptbp1 inactivation (initiated around E11.5 with the onset of Chx10-Cre activity) on gene expression and splicing, the data convincingly demonstrate that retinal cell type composition remains largely unaffected. This study is highly significant since it challenges the prevailing view of Ptbp1 as a central repressor of neurogenesis and highlights the need to further investigate, or re-evaluate, its role in other model systems and regions of the CNS.

Weaknesses:

A limitation of the study is the use of the Chx10-Cre driver, which initiates recombination around E11. This timing does not permit assessment of Ptbp1 function during the earliest phases of retinal development, if expressed at that time.

Comments on revisions:

The authors have thoroughly and satisfactorily addressed all my previous comments.

<https://doi.org/10.7554/eLife.108331.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

The researchers sought to determine whether Ptbp1, an RNA-binding protein formerly thought to be a master regulator of neuronal differentiation, is required for retinal neurogenesis and cell fate specification. They used a conditional knockout mouse line to remove Ptbp1 in retinal progenitors and analyzed the results using bulk RNA-seq, single-cell RNA-seq, immunohistochemistry, and EdU labeling. Their findings show that Ptbp1 deletion has no effect on retinal development, since no defects were found in retinal lamination, progenitor proliferation, or cell type composition. Although bulk RNA-seq indicated changes in RNA splicing and increased expression of late-stage progenitor and photoreceptor genes in the mutants, and single-cell RNA-seq detected relatively minor transcriptional shifts in Müller glia, the overall phenotypic impact was low. As a result, the authors conclude that Ptbp1 is not required for retinal neurogenesis and development, thus contradicting prior statements about its important role as a master regulator of neurogenesis. They argue for a reassessment of this stated role. While the findings are strong in the setting of the retina, the larger implications for other areas of the CNS require more investigation. Furthermore, questions about potential reimbursement from Ptbp2 warrant further research.

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This study calls into doubt the commonly held belief that Ptbp1 is a critical regulator of neurogenesis in the CNS, particularly in retinal development. The adoption of a conditional knockout mouse model provides a reliable way for eliminating Ptbp1 in retinal progenitors while avoiding the off-target effects often reported in RNAi experiments. The combination of bulk RNA-seq, scRNA-seq, and immunohistochemistry enables a thorough examination of molecular and cellular alterations at both embryonic and postnatal stages, which strengthens the study's findings. Furthermore, using publicly available RNA-Seq datasets for comparison improves the investigation of splicing and expression across tissues and cell types. The work is wellorganized, with informative

figure legends and supplemental data that clearly show no substantial phenotypic changes in retinal lamination, proliferation, or cell destiny, despite identified transcriptional and splicing modifications.

We thank the Reviewer for their evaluation of the strengths of the study.

Weaknesses:

The retina-specific method raises questions regarding whether Ptbp1 is required in other CNS locations where its neurogenic roles were first proposed. The claim that Ptbp1 is "fully dispensable" for retinal development may be toned down, given the transcriptional and splicing modifications identified. The possibility of subtle or transitory impacts, such as ectopic neuron development followed by cell death, is postulated, but not completely investigated. Furthermore, as the authors point out, the compensating potential of increased Ptbp2 warrants additional exploration. Although the study performs well in transcriptome and histological analyses, it lacks functional assessments (such as electrophysiological or behavioral testing) to determine if small changes in splicing or gene expression affect retinal function. While 864 splicing events have been found, the functional significance of these alterations, notably the 7% that are neuronal-enriched and the 35% that are rod-specific, has not been thoroughly investigated. The manuscript might be improved by describing how these splicing changes affect retinal development or function.

We have revised the text to address these points as requested.

Reviewer #2 (Public review):

Summary:

Ptbp1 has been proposed as a key regulator of neuronal fate through its role in repressing neurogenesis. In this study, the authors conditionally inactivated Ptbp1 in mouse retinal progenitor cells using the Chx10-Cre line. While RNA-seq analysis at E16 revealed some changes in gene expression, there were no significant alterations in retinal cell type composition, and only modest transcriptional changes in the mature retina, as assessed by immunofluorescence and scRNAseq. Based on these findings, the authors conclude that Ptbp1 is not essential for cell fate determination during retinal development.

Strengths:

Despite some effects of Ptbp1 inactivation (initiated around E11.5 with the onset of Chx10-Cre activity) on gene expression and splicing, the data convincingly demonstrate that retinal cell type composition remains largely unaffected. This study is highly significant since it challenges the prevailing view of Ptbp1 as a central repressor of neurogenesis and highlights the need to further investigate, or re-evaluate, its role in other model systems and regions of the CNS.

We thank the Reviewer for their evaluation of the strengths of the study.

Weaknesses:

A limitation of the study is the use of the Chx10-Cre driver, which initiates recombination around E11. This timing does not permit assessment of Ptbp1 function during the earliest phases of retinal development, if expressed at that time.

We have revised the text to address the potential limitations of the use of the Chx10-Cre driver in this study.

Reviewer #1 (Recommendations for the authors):

(1) The author only selected scRNA-Seq datasets to examine the expression patterns of Ptbp1 in the retina; incorporating immunostaining analysis in the mouse retina is necessary.

Ptbp1 expression patterns in the mouse retina were performed in Fig. 1b-1d, where Ptbp1 expression was analyzed via immunostaining for Ptbp1 protein in Chx10-Cre control and Ptbp1KO retinas at E14, P1, and P30, and are quantified in Fig. 1e.

(2) In Figure 1, Ptbp1 signals were still detected in the KO mice, with the author suggesting that this may indicate cross-reactivity with an unknown epitope. Why is this unknown epitope only detected in the ganglion cell layer? Additional antibodies are needed to confirm the staining results. Furthermore, it is essential to verify the KO at the mRNA level using PCR.

We are unsure of the identity of this cross-reacting epitope, although it might be Ptbp2, which is enriched expressed in immature retinal ganglion cells (Fig. S1). In any case, we do not believe that the identity of this epitope is not relevant to assessing the efficiency of Ptbp1 deletion, as it is not detectably expressed in retinal ganglion cells in any case (Fig. S1).

Although the heatmap in Figure 2B indicates a decrease in Ptbp1 levels in the KO mice, the absence of statistical data makes it difficult to evaluate the KO efficiency.

Respectfully, we believe that Ptbp1 knockout efficiency is adequately addressed using immunohistochemistry, and that further statistical analysis is not essential here.

Cre staining of the Chx10-Cre;Ptbp1lox/lox mice or using reporter lines is also suggested to indicate the theoretically knockout cells. Providing high-power images of the Ptbp1 staining would help readers clearly recognize the staining signals.

To clarify the identity of the knockout cells, we have updated Figure 1 to include the Chx10-CreEGFP staining which more clearly delineates the cells in which Ptbp1 is deleted. Regarding verification of the knockout, we believe additional PCR assays are not necessary, as we have already demonstrated efficient loss of Ptbp1 in Chx10-Cre-expressing cells at the RNA level by both single-cell RNA-sequencing and bulk RNA-sequencing, and also at the protein level by immunohistochemistry. Sun1-GFP Cre reporter lines are also used in Figures 1 and S2 to visualize patterns of Cre activity, a point which is now highlighted in the text. Together, these approaches provide sufficient evidence for effective Ptbp1 knockout.

(3) The possibility of ectopic neuron formation followed by cell death is intriguing but underexplored. Consider adding apoptosis assays (e.g., TUNEL staining) at early developmental stages to test this hypothesis.

While apoptosis assays such as TUNEL staining would be helpful to address this hypothesis, we feel incorporating these additional experiments is currently beyond the scope of this study. We agree the possibility of cell death is intriguing and plan to explore this in future work.

(4) On page 4, the statement "We did not observe any significant differences ... Chx10Cre;Ptbp1lox/lox mice (Fig. 2b,c)" should refer to Fig. 3b,c instead.

We have changed the text to refer to Fig. 3b,c.

(5) *The labeling in Figure 3 as "Cre-Ptbp1" is inconsistent with the figure legend "Ptbp1-Ctrl."*

This language was used because the samples for EdU staining in Figure 3 were Chx10-Cre negative Ptbp1^{lox/lox} mice. We have updated the language in the manuscript and figure to reflect the genotypes more clearly.

(6) *P30 mice are still sexually immature; the term "adolescent" or "juvenile" should be used instead of "adult."*

We have updated the language in the text from “adult” to “adolescent” to describe P30 mice, although the retina itself is mature by this age.

Reviewer #2 (Recommendations for the authors):

(1) *As mentioned in the public review, a limitation of the study is that Ptbp1 KO is not induced prior to E11. The authors should acknowledge this limitation and include in the Discussion that the use of the Chx10-Cre line does not permit evaluation of a potential role for Ptbp1 during very early stages of retinal development, should it be expressed at that time (an aspect that would be important to determine).*

We and have added this limitation to the Discussion in the sentence highlighted below.

Furthermore, the use of the Chx10-Cre transgene in this study does not exclude a potential role for Ptbp1 during very early stages of retinal development prior to E11 (pg. 6).

(2) *While the data convincingly show no significant changes in retinal cell type distribution in Ptbp1 mutants, the claims in the abstract and introduction that Ptbp1 is "dispensable for retinal development" or "dispensable for the process of neurogenesis" may be overstated. Indeed, the results indicate that loss of Ptbp1 function influences retinal development by promoting neurogenesis through induction of a neuronal-like splicing program in neural progenitors. Concluding solely that Ptbp1 is dispensable for retinal cell fate specification, rather than for retinal development as a whole, would thus seem more accurate.*

We have updated the language in the text to reflect Ptbp1's role in regulating retinal cell fate specification more clearly.

(3) *The authors conclude from Figure 5 that "No changes in the identity or composition of any retinal cell type were observed." Which statistical test was applied to support this conclusion? The figure indicates that Müller cells comprise 10.5% of the total cell population in controls versus 8.2% in Ptbp1-KO retinas. It may be important to consider the overall distribution of glia versus all neurons (rather than each neuron subtype individually). While the observed difference (~2% more glia at the expense of neurons) appears modest, it would be important to determine whether this trend is consistent and statistically significant.*

To evaluate cell type composition, we performed differential expression analysis across all major retinal cell types and compared proportional cell type representation between control and Ptbp1 KO retinas. While these analyses did not reveal marked differences in any specific cell type, we acknowledge that the scRNA-Seq dataset includes a single experimental replicate, containing two retinas in each replicate. Therefore, we cannot draw firm statistical

conclusions regarding the relative distribution of glia versus neurons, and the modest difference observed in glia cell proportion should be interpreted with caution. We agree that assessing glia-to-neuron ratios across additional replicates will be important in future studies.

(4) Referring to Figure S1 (scRNA-seq data), the authors state that Ptbp1 mRNA is robustly expressed in retinal progenitors and Müller glia in both mouse and human retina. While the immunostaining in Figure 4 indeed clearly shows strong expression in Müller cells, the scRNAseq data presented in Figure S1 do not support the claim of "robust" expression in Müller glia in the mouse retina. This is even more striking in the human data, where panels F and H show that Ptbp1 is expressed at extremely low, certainly not "robust", levels in Müller cells. The corresponding sentence in the Results section should therefore be revised to more accurately reflect the data presented in Figure S1, or be supported by complementary immunofluorescence evidence.

We thank the reviewer for this comment. We have revised this section of the Results to better reflect Fig S1, as follows:

We observe high expression levels of Ptbp1 mRNA in primary retinal progenitors in both species and Müller glia in mouse retina, with weaker expression in neurogenic progenitors, and little expression detectable in neurons at any developmental age.

(5) When mentioning potential compensation by Ptbp2, the authors may also consider discussing the possibility that compensatory mechanisms can differ between knockdown and knockout approaches. In this context, it is noteworthy that a recent study by Konar et al., Exp Eye Res, 2025 (published after the submission of the present manuscript) reports that Ptbp1 knockdown promotes Müller glia proliferation in zebrafish.

We thank the reviewer for this suggestion. To address this, we have included a section considering this possibility in the discussion section highlighted below.

It is also possible that compensatory mechanisms differ between knockdown and knockout approaches. Notably, a recent study (Konar et al. 2025) reported that Ptbp1 knockdown promotes Müller glia proliferation in zebrafish, suggesting that effects of acute reduction of Ptbp1 may not fully mirror those of complete loss-of-function.

(6) The statistical analyses were performed using a t-test. However, this parametric test is not appropriate for experiments with low sample sizes. A non-parametric test, such as the MannWhitney test, would be more suitable in this context. Furthermore, performing statistical analysis on $n = 2$ (Figure 3C) is not statistically valid.

We thank the reviewer for this comment. We agree that with a small n , non-parametric tests are more appropriate. We have added additional retinas (now $n=5$) for the Ptbp1-KO condition in Figure 3C and reanalyzed with the appropriate non-parametric Mann-Whitney test. For all other datasets with sufficient replicates ($n \geq 4/\text{genotype}$), parametric tests such as unpaired t-tests remain valid, and the results are consistent with non-parametric testing.

(7) Figure S3 is accompanied by only a brief explanation in the Results section (a single sentence despite the figure containing six panels), which makes it difficult for readers unfamiliar with this type of data to interpret.

We thank the reviewer for the suggestion. To address this, we have included a more detailed explanation of Supplementary Figure S3 to better clarify our analysis of mature neuronal and glial cell types in both Ptbp1-deficient and wild-type animals. The relevant text now reads:

Notably, splicing patterns in *Ptbp1*-deficient retinas showed stronger correlation with *Thy1*-positive neurons—which exhibit low *Ptbp1* expression—and minimal overlap with microglia and auditory hair cells, the adult cell types with the highest *Ptbp1* levels (Fig. S3).

Gene expression and splicing changes were compared across several reference tissues: heart tissue and *Thy1*-positive neurons, mature hair cells, microglia, and astrocytes (Fig. S3a,b). A heatmap of differentially expressed genes showed that while *Ptbp1*-deficient retinas diverged from WT retinas, their expression profiles did not resemble those of fully differentiated cell types like rods, astrocytes, or adult WT retina (Fig. S3c). Consistently, Pearson correlation analysis revealed that *Ptbp1*-deficient and WT retinas were more similar to each other than to fully differentiated neuronal or glial populations (Fig. S3d). Splicing profile analysis further revealed that while there was high correlation of PSI between *Ptbp1*-deficient and WT retinas, *Ptbp1*-deficient retinas more closely resembled *Thy1*-positive neurons, whereas WT retinas aligned more strongly with mature cells such as astrocytes, microglia, and auditory hair cells (Fig. S3e,f). Together, these results suggest that although *Ptbp1* loss induces hundreds of alternative splicing events, the magnitude of PSI changes in the KO retinas remains considerably lower than that seen in fully differentiated cell types (Extended Data 3). Thus, while a subset of splicing events overlaps with those characteristic of mature neurons or rods, the overall splicing and expression profiles of KO retinas are more similar to those of developing retinal tissue rather than terminally differentiated neuronal or glial populations.

(8) To assess progenitor proliferation, the authors performed EdU labeling experiments in P0 retinas. Is there a rationale for not examining earlier developmental time points to evaluate potential effects on early RPCs?

We thank the reviewer for this comment. We chose to perform EdU labeling experiments at P0 for several reasons. P0 represents a developmental stage where RPCs are actively proliferating and represent ~35% of all retina cells, and the retina is transitioning to intermediate-late-stage development, providing sufficient time to ensure efficient and widespread disruption of *Ptbp1*. Earlier embryonic timepoints were not examined here, as addressing all stages of development was beyond the scope of this current study. However, we agree that investigating whether *Ptbp1* plays stage-specific roles during development on early RPCs is an important question and potential future direction.

*(9) In Figure S2, panel D shows staining in GCL under the *Ptbp1* condition that does not make sense and is inconsistent with panel C. If possible, the authors should provide an alternative image to prevent any confusion.*

Thank you for bringing this to our attention. The image shown for *Ptbp1*-KO in Figure 2d shows Sun1-eGFP labeling, which labels every cell affected by the Cre condition. The genotype for this mouse was *Chx10-Cre;Ptbp1lox/lox;Sun1-GFP*. We apologize for any confusion and have updated the genotype in the figure legend.

(10) The authors should revise the following sentence at the end of the Discussion section, as its meaning is unclear: "...and conditions for in vitro analysis may have accurately replicated conditions in the native CNS."

We thank the reviewer for this comment and have revised this sentence in the discussion for the sentence below.

Previous studies using knockdown may have been complicated by off-target effects (Jackson et al. 2003), and conditions for in vitro analysis may not have accurately replicated conditions in the native CNS.

<https://doi.org/10.7554/eLife.108331.2.sa0>